

Tensor computations

Frobenius inner products minimize the algebraic complexity of rank-one approximation

Difficulty: challenging **Importance:** interesting to the community

Statement

Fix integers $k \geq 1$, $n_j \geq 2$, $d_j \geq 1$, with $\sum_j d_j \geq 3$. Set

$$W_{\mathbb{R}} = \bigotimes_{j=1}^k \text{Sym}^{d_j}(\mathbb{R}^{n_j}), \quad X = \{c v_1^{\otimes d_1} \otimes \cdots \otimes v_k^{\otimes d_k} : c \in \mathbb{C}, v_j \in \mathbb{C}^{n_j}\} \subset W_{\mathbb{C}}.$$

For a positive definite symmetric real bilinear form Q on $W_{\mathbb{R}}$, extend it complex-bilinearly. Define $\text{ED}_Q(X)$ to be the number of complex critical points on the smooth nonzero part of X of $Z \mapsto Q(A - Z, A - Z)$, for Zariski-generic $A \in W_{\mathbb{C}}$, with algebraic multiplicities.

Let Q_F be the restriction of the entrywise Frobenius inner product from the full tensor product, with standard Euclidean inner products on the factors. Is

$$\text{ED}_Q(X) \geq \text{ED}_{Q_F}(X)$$

true for every such Q ? The bilinear complexification uses no complex conjugates; the Frobenius restriction includes the usual repeated-entry weights on symmetric tensors.

Relevance

The ED degree counts algebraic stationary solutions of weighted rank-one approximation. The conjecture says the natural tensor metric minimizes this complexity, across arbitrary positive definite weights and partial symmetries.

References

1. K. Kozhasov, A. Muniz, Y. Qi, and L. Sodomaco, *On the minimal algebraic complexity of the rank-one approximation problem for general inner products*. [Primary text](#), Conjecture 3.9; Theorem 3.12 and §4; [publication DOI](#).
2. J. Draisma, E. Horobeț, G. Ottaviani, B. Sturmfels, and R. R. Thomas, *The Euclidean distance degree of an algebraic variety*, Found. Comput. Math. (2016). [Primary preprint](#), §§1–2 for ED degree and §4 for average ED degree.

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The October 13, 2025 revision of reference 1 explicitly retains Conjecture 3.9. It proves local minimality for all these varieties and global minimality for matrices, binary symmetric tensors, and ternary symmetric cubics. Searches "Frobenius" "ED degree" "conjecture" "2026" and the exact paper title with *proof* located no general resolution. This bounded check distinguishes global minimality from the proved local theorem.